	Raffles Institution Physics Department Revision Package 3	Temperature and Ideal Gases	12

***Important note:** *denotes questions that may require knowledge from later chapters in order to solve them.*

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/14]

An ideal gas is contained in a metal box. Which is not true?

A Increasing the temperature of the gas will increase the rate of collision of the gas molecules with the wall.

B Decreasing the internal energy of the gas will decrease the change of momentum of the gas molecules when it hit the walls.

C Increasing the dimensions of the box will increase the frequency of collision of the gas molecules with the walls.

D The speeds of the gas molecules are different but the pressure of the gas on the walls for the same temperature is constant.

2. [AJC/H2/Prelims 2018/P1/11]

Which statement explains why the thermodynamic temperature scale is considered more fundamental than the centigrade temperature scale?

A The thermodynamic temperature scale is determined using the triple point of water instead of ice point, which is more reproducible.

B The thermodynamic temperature scale is independent of the properties of materials.

C The thermodynamic temperature scale can measure a greater range of temperatures.

D The thermodynamic temperature scale is related to the random kinetic energy of the particles.

3. [CJC/H2/Prelims 2018/P1/10]

The density of a sample of helium gas at the pressure of 100 kPa is $0.180\ \text{kg m}^{-3}$. The root-mean-square speed of the helium molecules is

A $41\ \text{m s}^{-1}$ B $561\ \text{m s}^{-1}$ C $1290\ \text{m s}^{-1}$ D $1685\ \text{m s}^{-1}$

4. [CJC/H2/Prelims 2018/P1/11]

A car tyre, initially at 28 °C, has been inflated to a pressure of 160 kPa as indicated by the pressure gauge. This means that the pressure in the tyre is 160 kPa above the atmospheric pressure of 100 kPa.

After driving on hot roads, the temperature of the air in the tyre is 65 °C.

What is the percentage increase in the pressure gauge reading?

A 10% B 20% C 200% D 270%

5. [DHS/H2/Prelims 2018/P1/12]

In deriving the equation $pV = \frac{1}{3}Nm\langle c^2\rangle$ in the simple kinetic theory of gases, which of the following is **not** taken as a valid assumption?

A The molecules suffer negligible change of momentum on collision with the walls of the container.

B Collisions with the walls of the container and with other molecules cause no change in the average kinetic energy of the molecules.

C The duration of a collision is negligible compared with the time between collisions.

D The volume of the molecules is negligible compared with the volume of the gas.

6. [DHS/H2/Prelims 2018/P1/13]

Atoms of neon are at a temperature such that the root mean square (r.m.s.) speed of its atoms is $400\ \text{m s}^{-1}$.

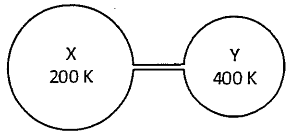
What will be the r.m.s. speed of molecules of hydrogen at the same temperature?

Mass of neon atom = 20 *u*; mass of hydrogen molecule = 2 *u*.

A $130\ \text{m s}^{-1}$ B $400\ \text{m s}^{-1}$ C $1300\ \text{m s}^{-1}$ D $4000\ \text{m s}^{-1}$

7. [EJC/H2/Prelims 2018/P1/8]

In the diagram below, the volume of bulb X is twice that of bulb Y. The system is filled with an ideal gas and a steady state is established with the bulbs held at 200 K and 400 K.



There are *b* moles of gas in X. How many moles of gas are there in Y?

A $b/4$ B $b/2$ C b D $2b$

8. [HCI/H2/Prelims 2018/P1/16]

Two vessels X and Y contain ideal gases at the same temperature *T*. The pressures of ideal gases in X and Y are *P* and $P/4$ respectively. The volume of X is 1.5 times that of Y. The vessels are connected by a narrow tube with a tap. The tap is initially closed. The temperature of the gas is maintained at the constant temperature *T*.

What is the pressure of the gas at equilibrium when the tap is opened?

A 0.50 *P* B 0.70 *P* C 0.75 *P* D 1.50 *P*

9. [JJC/H2/Prelims 2018/P1/11]

The r.m.s. speed of the molecules of a gas at 295 K is decreased by 20 %.

What is the new temperature of the gas?

A –84.4 °C

B –37.2 °C

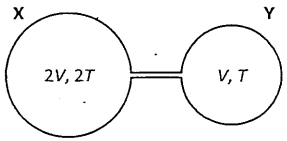
C 189 °C

D 236 °C

10. **[JJC/H2/Prelims 2018/P1/13]**

An ideal gas is contained in two spherical containers X and Y of volume $2V$ and V respectively, connected by a hollow tube of negligible volume. The containers X and Y are maintained at temperatures $2T$ and T respectively.

The setup is shown in the figure below.



Determine the ratio $\dfrac{\text{number of moles of gas in container X}}{\text{number of moles of gas in container Y}}$.

A 0.25

B 1

C 2

D 4

11. **[MI/H2/Prelims 2018/P1/14]**

The mean kinetic energy of the gas molecules of an ideal gas at 573 K is 1.6×10^{-23} J.

Which of the following correctly shows the values of temperature and mean kinetic energy of the gas molecules when the gas molecules are travelling twice as fast?

Which of the following could be correct?

	temperature/ K	kinetic energy/ J
A	873	3.2×10^{-23}
B	1370	3.2×10^{-23}
C	2292	6.4×10^{-23}
D	2573	6.4×10^{-23}

12. **[MJC/H2/Prelims 2018/P1/14]**

An ideal gas in a container of fixed volume 1.0 m^3 has a pressure of 3.0×10^5 Pa at a temperature of 200 K. The gas is heated until the temperature reaches 400 K. Some gas is released from the container during the heating to keep the pressure constant.

What volume does the gas released from the container occupy, if it is at atmospheric pressure of 1.0×10^5 Pa and at a room temperature of 300 K?

A 0.500 m^3 B 2.00 m^3 C 2.25 m^3 D 4.50 m^3

13. **[MJC/H2/Prelims 2018/P1/16]**

The molecules of an ideal gas at thermodynamic temperature T have a root-mean-square speed c.

The gas is heated to temperature $2T$.

What is the new root-mean-square speed of the molecules?

A $\sqrt{2}c$ B $2\sqrt{2}c$ C $2c$ D $4c$

14. **[PJC/H2/Prelims 2018/P1/10]**

A gas cylinder has volume of 4.0×10^{-4} m^3. It contains gas at a temperature of 300 K and a pressure of 500 kPa. A worker fills the cylinder to a pressure of 620 kPa without changing the temperature.

What is the amount of gas in number of moles that must be pumped into the cylinder?

A 0.019 mol

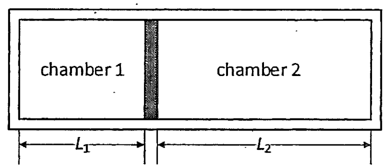
B 0.080 mol

C 0.10 mol

D 0.18 mol

15. **[RVHS/H2/Prelims 2018/P1/8]**

A closed cylinder has a freely moving piston separating chambers 1 and 2. Chamber 1 contains 20 mg of nitrogen gas and chamber 2 contains 45 mg of helium gas. The relative molecular mass of nitrogen is 28 and that of helium is 4.



When both pressure and thermal equilibrium are established between chamber 1 and 2, what is the ratio L_1/L_2 ?

A 0.063 B 0.44 C 2.3 D 16

16. **[SRJC/H2/Prelims 2018/P1/15]**

In a mixture of two monatomic ideal gaseous X and Y, the molecules of Y have thrice the mass of those of X. The mixture is in thermal equilibrium and the molecules of Y have a mean translational kinetic energy of E_k.

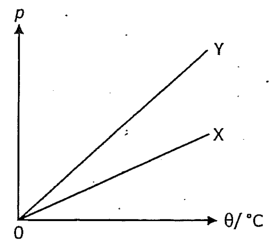
What is the mean translational kinetic energy of the molecules of X?

A $\frac{1}{3}E_k$ B $\frac{1}{2}E_k$ C E_k D $3E_k$

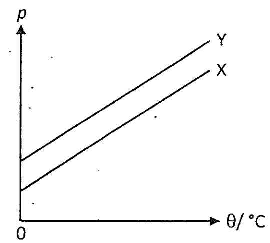
17. **[TPJC/H2/Prelims 2018/P1/16]**

Two closed vessels X and Y contain equal masses of an ideal gas. X has a greater volume than Y. When the temperature changes, which of the following represents the variation of the pressure p (in Pa) of the gas in each vessel with temperature θ (in °C)?

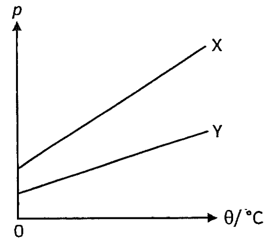
A



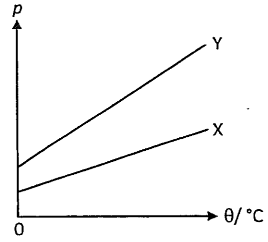
B



C



D



18. [VJC/H2/Prelims 2018/P1/11]

The density of helium at 150 kPa is 0.178 kg m^{-3}.

What is the root-mean-square speed of its particles?

A 130 m s^{-1} B 232 m s^{-1}

C 1300 m s^{-1} D 1600 m s^{-1}

19. [YJC/H2/Prelims 2018/P1/11]

A flask of volume 3.0×10^{-4} m^3 contains nitrogen gas at a pressure of 2.5×10^5 Pa and temperature of 27 °C.

What is the number of nitrogen molecules in the flask?

A 0.030 B 0.25 C 1.8×10^{22} D 2.0×10^{23}

20. [YJC/H2/Prelims 2018/P1/12]

Which of the following statements is not a valid assumption of kinetic theory of gases?

A Intermolecular forces between molecules are negligible.

B The molecules experience negligible change in momentum on collision with the walls of the container.

C The molecules experience negligible change in average kinetic energy on collision with the walls of the container.

D The duration of a collision is negligible compared with time between collisions.

II. Structured Questions (ACJC – YJC)

1. [AJC/H2/Prelims 2018/P2/3]

a. i. Explain the concept of *absolute zero* on the thermodynamic temperature scale. [1]

ii. Explain how empirical evidence leads to the concept of *absolute zero*. [3]

b. Two insulated gas cylinders A and B are connected by a tube of negligible volume, as shown in Fig. 1.1.

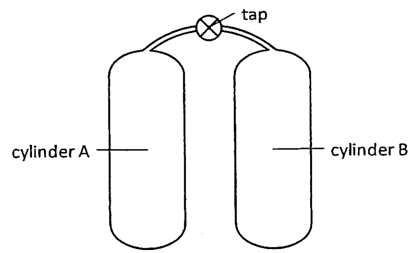


Fig. 1.1

Each cylinder has an internal volume of 2.0×10^{-2} m^3. Initially, the tap is closed and cylinder A contains 1.2 mol of an ideal gas at a temperature of 37 °C. Cylinder B contains the same ideal gas at pressure 1.2×10^5 Pa and temperature 37 °C.

The tap is opened and some gas flows from cylinder A to cylinder B. Using the fact that the total amount of gas is constant, determine the final pressure of the gas in the cylinders. [4]

[Total: 8]

2. [IJC/H2/Prelims 2018/P3/3]

a. An ideal gas is assumed to consist of atoms or molecules that behaves as hard, identical spheres that are in continuous motion and undergo elastic collisions

State two further assumptions of the kinetic theory of gases. [2]

b. Helium-4 (^4_2He) may be assumed to be an ideal gas.

i. Show that the mass of one atom of helium-4 is 6.6×10^{-27} kg. [1]

ii. Determine the internal energy of 1.2 mol of helium-4 gas at a temperature of 27 °C. [1]

iii. Determine the root-mean-square (r.m.s.) speed of a helium-4 atom at a temperature of 27 °C. [2]

iv. The above r.m.s. speed for helium is less than the escape speed for particles on the Earth's surface and yet there are still some helium atoms that escape the Earth's atmosphere. [1]

[Total: 7]

3. [JJC/H2/Prelims 2018/P3/8]

a. i. State two assumptions of the kinetic theory of gases. [2]

ii. A sealed container holds a mixture of nitrogen-14 and helium-4 gas molecules at temperature of 290 K.

1. Determine the average kinetic energy of a helium molecule. [2]

2. Explain whether it is possible for the mean square speed of the nitrogen molecules to be different from the helium molecules at the same temperature. [1]

4. **[MI/H2/Prelims 2018/P3/3 Part]**

   a. i. One of the assumptions of the kinetic theory of gases is that the motion of gas molecules is random. Therefore explain why the average momentum of the gas molecules in a container is zero. **[2]**

   ii. According to the kinetic theory of gases, explain how a gas exerts a pressure on the walls of a container. **[3]**

[Total: 5]

5. **[NYJC/H2/Prelims 2018/P3/5 Part]**

   a. *State what is meant by the term *internal energy of an ideal gas*. **[1]**

   b. The pressure p exerted by an ideal gas is given by the equation

$$p=\frac{1}{3}\rho\langle c^2\rangle$$

   State what the symbols ρ and $\langle c^2\rangle$ represent. **[2]**

   c. Use the equation given in **(b)** to show that the total kinetic energy of the molecules of an ideal gas of volume V and pressure p is given by $\frac{3}{2}pV$. **[2]**

[Total: 5]

6. **[PJC/H2/Prelims 2018/P3/3]**

   a. State two basic assumptions of the kinetic theory of gases. **[2]**

   b. A cylinder contains 2.5 mol of an ideal monatomic gas at a pressure of 1.8×10^5 Pa and temperature of 288 K. The gas has density of 2.7 kg m^{-3}.

   i. Determine the root-mean-square (r.m.s.) speed of the gas. **[1]**

   ii. *The gas is heated at constant pressure until its volume is doubled. Calculate

   1. the increase in internal energy of the gas, **[2]**
   2. the work done on the gas, **[2]**
   3. the thermal energy supplied to the gas. **[2]**

[Total: 9]

7. **[YJC/H2/Prelims 2018/P2/3 Modified]**

   a. A cylinder of fixed volume contains some nitrogen gas at a pressure of 2.10×10^5 Pa. The gas has a density 1.50 kg m^{-3} and the molar mass of nitrogen is 28.0 g, calculate

   i. the root-mean-square speed of the nitrogen molecule, **[2]**

   ii. the temperature of the nitrogen gas in the cylinder. **[2]**

   iii. Using kinetic theory of gases, state and explain what will happen to the pressure of the nitrogen gas when the temperature of the gas is increased. **[2]**

[Total: 6]